



CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 15: Similarity (With Applications to Maps and Models)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Similar triangles

1. $\triangle ABC \sim \triangle EFG$ with $\angle ABC = \angle EFG = 60^\circ$, $AB = 15$ cm, $BC = 3$ cm and $EF = 75$ cm. Find FG .
(a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm **ICSE 2025, 1 mark**

SOLUTION

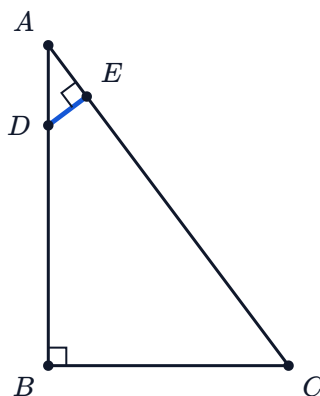
Corresponding sides are proportional: $\frac{AB}{EF} = \frac{BC}{FG}$.

$$\frac{15}{75} = \frac{3}{FG}, \text{ so } FG = 15 \text{ cm.}$$

Answer: (a)

2 Maps and models

2. In $\triangle ABC$, $\angle ABC = 90^\circ$, $AB = 20$ cm and $AC = 25$ cm. D is a point on AB and E is a point on AC with $DE \perp AC$ and $DE = 3$ cm. (a) Prove that $\triangle ABC \sim \triangle AED$. (b) Find BC , AD and AE . (c) $BCED$ is a plot of land drawn on a map, and its actual area is 576 m^2 . Find the scale factor of the map. **ICSE 2025, 4 marks**



SOLUTION

(a) In $\triangle ABC$ and $\triangle AED$: $\angle A$ is common and $\angle ABC = \angle AED = 90^\circ$. So $\triangle ABC \sim \triangle AED$ (AA).

(b) $BC = \sqrt{25^2 - 20^2} = 15$ cm. From the similar triangles, $\frac{AB}{AE} = \frac{BC}{ED} = \frac{AC}{AD} = \frac{15}{3} = 5$.
 $AE = \frac{20}{5} = 4$ cm and $AD = \frac{25}{5} = 5$ cm.

(c) Area of $BCED$ on the map = area of $\triangle ABC$ – area of $\triangle AED = \frac{1}{2} \times 20 \times 15 - \frac{1}{2} \times 4 \times 3 = 150 - 6 = 144 \text{ cm}^2$.

Actual area = $576 \text{ m}^2 = 5760000 \text{ cm}^2$. $k^2 = \frac{144}{5760000} = \frac{1}{40000}$, so $k = \frac{1}{200}$.

Answer: (b) $BC = 15$ cm, $AD = 5$ cm, $AE = 4$ cm (c) scale factor $1 : 200$

3 Conditions for similarity

3. $\triangle ABC \sim \triangle PQR$. AD bisects $\angle BAC$ and meets BC at D ; PS bisects $\angle QPR$ and meets QR at S . Which statement is true? (a) $\triangle ABC \sim \triangle PQS$ (b) $\triangle ABD \sim \triangle PQS$ (c) $\triangle ABD \sim \triangle PSR$ (d) $\triangle ABC \sim \triangle PSR$ **ICSE 2024, 1 mark**

SOLUTION

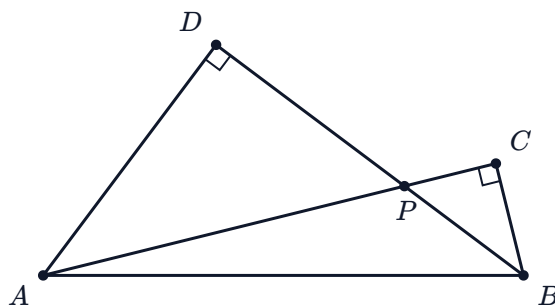
$\angle B = \angle Q$ and $\angle A = \angle P$ (similar triangles), so $\angle BAD = \frac{1}{2} \angle A = \frac{1}{2} \angle P = \angle QPS$.

Two angles of $\triangle ABD$ equal two angles of $\triangle PQS$.

Answer: (b)

4 Areas of similar triangles

4. $\triangle ADB$ and $\triangle ACB$ are right-angled at D and C , on the same side of AB , and AC meets BD at P . $AB = 10$ cm, $AD = 6$ cm, $BC = 2.4$ cm and $DP = 4.5$ cm. (a) Prove that $\triangle APD \sim \triangle BPC$. (b) Find BD and PB . (c) Hence find PA . (d) Find area of $\triangle APD$: area of $\triangle BPC$. **ICSE 2024, 4 marks**



SOLUTION

(a) $\angle ADP = \angle BCP = 90^\circ$ and $\angle APD = \angle BPC$ (vertically opposite). So $\triangle APD \sim \triangle BPC$ (AA).

(b) In $\triangle ADB$: $BD = \sqrt{10^2 - 6^2} = 8$ cm. $PB = BD - DP = 8 - 4.5 = 3.5$ cm.

(c) From the similar triangles, $\frac{PA}{PB} = \frac{AD}{BC} = \frac{6}{2.4} = 2.5$, so $PA = 2.5 \times 3.5 = 8.75$ cm.

(d) Ratio of areas = $\left(\frac{AD}{BC}\right)^2 = (2.5)^2 = \frac{25}{4}$

Answer: (b) $BD = 8$ cm, $PB = 3.5$ cm (c) $PA = 8.75$ cm (d) $25 : 4$

Note: the measurements in this question do not fit together exactly (the Pythagoras theorem in $\triangle ADP$ would give $PA = 7.5$ cm). The word "hence" tells you to use the similar triangles, as done here.

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